Fundamentals of Linear Algebra for AI

Invertibility

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Table of Contents (I/V)

Invertibility

The Inverse Matrix ($\mathbf{A}^{-1}$)

- **The Goal:** The matrix inverse is the mathematical operation that undoes matrix multiplication.
- **Analogy:** It is the matrix equivalent of division in ordinary algebra (multiplying by $\mathbf{A}^{-1}$ is like dividing by $\mathbf{A}$).
- **Mathematical Definition:** If $\mathbf{A}$ is a square matrix, its inverse $\mathbf{A}^{-1}$ satisfies:

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

- If $\mathbf{A}^{-1}$ exists, the unique solution to $\mathbf{A}\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

The Identity Matrix ($\mathbf{I}$)

The identity matrix is the multiplicative neutral element in matrix algebra, acting as "1" does in scalar algebra.

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2. Conditions for the Inverse to Exist (Invertibility)

- Not all matrices have an inverse. A matrix that possesses an inverse is called **invertible** or **non-singular**.

1. **The Dimension Condition: It Must Be Square**
   - The number of rows must equal the number of columns (e.g., 3×3).
2. **The Determinant Condition: Its Determinant Must Be Non-Zero**
   - This is the most crucial condition: $\det(\mathbf{A}) \neq 0$.
   - **Geometric Interpretation:** If $\det(\mathbf{A}) = 0$, the matrix collapses the space (e.g., transforms 3D space onto a 2D plane). This kind of transformation cannot be undone.

Calculation Methods (2x2)

3. Calculating the Inverse: The 2×2 Case

The 2×2 matrix is the simplest case and has a direct formula:

- If the matrix $\mathbf{A}$ is:

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

- Then its inverse $\mathbf{A}^{-1}$ is given by the formula:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

- **Recall:** The determinant $\det(\mathbf{A}) = ad - bc$.

Example

Let $\mathbf{A} = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$.

- $\det(\mathbf{A}) = (4)(6) - (7)(2) = 24 - 14 = 10$.
- $\mathbf{A}^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$.

Calculation Methods (Larger matrices)

4. Calculating the Inverse: The Gauss-Jordan Method

- This method is used for larger matrices (3×3 and up) and is generally the most computationally effective.
- **The Idea:** Apply Elementary Row Operations (EROs) to transform the matrix into the Identity Matrix.

1. **Setup:** Write the matrix $\mathbf{A}$ next to the Identity Matrix $\mathbf{I}$ in an augmented matrix form:

$$[\mathbf{A}|\mathbf{I}]$$

2. **Operations:** Apply EROs (scaling, addition, swapping rows) across the entire augmented matrix.
3. **Result:** Continue the operations until the left side is transformed into $\mathbf{I}$. The resulting right side is the inverse $\mathbf{A}^{-1}$:

$$\xrightarrow{\text{Elementary Row Operations}} \left[\mathbf{I}|\mathbf{A}^{-1}\right]$$

1. Introduction to Gauss-Jordan Elimination

- **What it is:** A systematic algorithm to solve systems of linear equations or find the inverse of a matrix.
- **Core Idea:** Transform an augmented matrix into **Reduced Row Echelon Form (RREF)** using Elementary Row Operations (EROs).
- **Why RREF?** In RREF, the solution to the system (or the inverse matrix) can be read directly from the augmented matrix.

Key Concepts Involved

- **Augmented Matrix:** Combines the coefficient matrix with the constant vector (e.g., $[\mathbf{A}|\mathbf{b}]$).
- **Elementary Row Operations (EROs):** The allowed manipulations:
  1. Swapping two rows.
  2. Multiplying a row by a non-zero scalar.
  3. Adding a multiple of one row to another row.

2. General Steps of Gauss-Jordan Elimination

1. **Form the Augmented Matrix:**
   - For solving $\mathbf{Ax} = \mathbf{b}$: Create $[\mathbf{A}|\mathbf{b}]$.
   - For finding $\mathbf{A}^{-1}$: Create $[\mathbf{A}|\mathbf{I}]$, where $\mathbf{I}$ is the identity matrix.
2. **Achieve Row Echelon Form (REF):**
   - Use EROs to get leading 1s (pivots) in each non-zero row.
   - Ensure all entries below a pivot are zero.
   - Ensure each leading 1 is to the right of the leading 1 in the row above it.
3. **Achieve Reduced Row Echelon Form (RREF):**
   - After REF, use EROs to make all entries above each pivot also zero.
   - (This is the "Jordan" step, as Gaussian Elimination only goes to REF).
4. **Read the Result:**
   - If solving $\mathbf{Ax} = \mathbf{b}$, the right side of $[\mathbf{I}|\mathbf{x}]$ is the unique solution $\mathbf{x}$.
   - If finding $\mathbf{A}^{-1}$, the right side of $[\mathbf{I}|\mathbf{A}^{-1}]$ is the inverse matrix.

Numerical example (1)

The System

We want to solve the following system:

$$x + 2y - z = 2 \tag{1}$$
$$3x + 8y + 2z = 11 \tag{2}$$
$$4x + 9y - z = 9 \tag{3}$$

- **Augmented Matrix $[\mathbf{A}|\mathbf{b}]$:** $\begin{pmatrix} \mathbf{1} & 2 & -1 & | & 2 \\ 3 & 8 & 2 & | & 11 \\ 4 & 9 & -1 & | & 9 \end{pmatrix}$
- **Operation 1:** Eliminate the 3 and 4 in Column 1.

$$R_2 \leftarrow R_2 - 3R_1$$
$$R_3 \leftarrow R_3 - 4R_1$$

Initial Setup and Forward Elimination

- **Result (First Pivot Set):**

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & \mathbf{2} & 5 & 5 \\ 0 & 1 & 3 & 1 \end{array}\right)$$

Completing Elimination (Row Echelon Form)

- **Operation 2:** Get a '1' in the pivot position $(2, 2)$. Swap or scale. We'll swap for simplicity.

$$R_2 \leftrightarrow R_3$$

- **Result (Pivot 2 Set):**

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & \mathbf{1} & 3 & 1 \\ 0 & 2 & 5 & 5 \end{array}\right)$$

- **Operation 3:** Eliminate the 2 below the new pivot.

$$R_3 \leftarrow R_3 - 2R_2$$

- **Result (Row Echelon Form - REF):**

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & \textbf{-1} & 3 \end{array}\right)$$

Backward Elimination (Reduced Row Echelon Form)

- **Operation 4:** Normalize the third pivot. $R_3 \leftarrow -1R_3$
- **Result (Pivots are all 1s):** $\begin{pmatrix} 1 & 2 & -1 & | & 2 \\ 0 & 1 & 3 & | & 1 \\ 0 & 0 & \mathbf{1} & | & -3 \end{pmatrix}$
- **Operation 5:** Eliminate above the third pivot.

$$R_1 \leftarrow R_1 + R_3$$
$$R_2 \leftarrow R_2 - 3R_3$$

- **Result (Final RREF):** $\begin{pmatrix} \mathbf{1} & 2 & 0 & | & -1 \\ 0 & \mathbf{1} & 0 & | & 10 \\ 0 & 0 & \mathbf{1} & | & -3 \end{pmatrix}$

Solution and Final RREF

Final Solution

The RREF matrix $[\mathbf{I}|\mathbf{x}]$ gives the solution vector $\mathbf{x}$:

$$\begin{pmatrix} 1 & 0 & 0 & | & -\mathbf{21} \\ 0 & 1 & 0 & | & \mathbf{10} \\ 0 & 0 & 1 & | & -\mathbf{3} \end{pmatrix}$$

Thus, the unique solution is:

$$x = -21, \quad y = 10, \quad z = -3$$

Solution verification

Proof by Matrix-Vector Multiplication

To verify, we check if $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 2 & -1 \\ 3 & 8 & 2 \\ 4 & 9 & -1 \end{pmatrix} \begin{pmatrix} -21 \\ 10 \\ -3 \end{pmatrix} = \begin{pmatrix} (1)(-21) + (2)(10) + (-1)(-3) \\ (3)(-21) + (8)(10) + (2)(-3) \\ (4)(-21) + (9)(10) + (-1)(-3) \end{pmatrix}$$

Verification of the Solution

The Final Check

Continuing the matrix multiplication:

$$\mathbf{Ax} = \begin{pmatrix} -21 + 20 + 3 \\ -63 + 80 - 6 \\ -84 + 90 + 3 \end{pmatrix} = \begin{pmatrix} \mathbf{2} \\ \mathbf{11} \\ \mathbf{9} \end{pmatrix}$$

Conclusion

Since the result $\begin{pmatrix} 2 \\ 11 \\ 9 \end{pmatrix}$ matches the original right-hand side vector $\mathbf{b}$, the solution is verified and the **Gauss-Jordan Elimination** was performed correctly.

Numerical Example (2)

Find the inverse of next matrix

$$\mathbf{Ax} = \begin{pmatrix} 2 & 2 \\ 4 & 5 \end{pmatrix}$$

Solution

- $det(A) = (2 \times 5) - (2 \times 4) = 2 > 0$
- $\begin{pmatrix} 2 & 2 & | & 1 & 0 \\ 4 & 5 & | & 0 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 2 & 2 & | & 1 & 0 \\ 0 & 1 & | & -2 & 1 \end{pmatrix} \xrightarrow{R_1 - 2R_2}$
  $\begin{pmatrix} 2 & 0 & | & 5 & -2 \\ 0 & 1 & | & -2 & 1 \end{pmatrix} \xrightarrow{\frac{1}{2}R_1} \begin{pmatrix} 1 & 0 & | & 2.5 & -1 \\ 0 & 1 & | & -2 & 1 \end{pmatrix}$

verification

$$\mathbf{Ax} = \begin{pmatrix} 2 & 2 \\ 4 & 5 \end{pmatrix} \times \begin{pmatrix} 2.5 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Numerical Example (3)

Find the inverse of next matrix

$$\mathbf{Ax} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Solution

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-7R_1]{R_2-4R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & -6 & -12 & -7 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3-2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array}\right].$$

verification

Since we have row-reduced the matrix on the left so that it has a zero row, we conclude that it is not invertible (dependency).

Numerical Example (4)

Find the inverse of next matrix

$$\mathbf{Ax} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$$

Solution

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-R_1]{R_2-R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array}\right] \xrightarrow[R_3-2R_2]{R_1-R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array}\right] \xrightarrow[R_2-3R_3]{R_1+2R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5/2 & 4 & -3/2 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array}\right]$$

Numerical Example (5)

The System and Initial Augmented Matrix

Consider the system:

$$\begin{aligned} x - 3y &= -5 \\ -2x + 7y &= 12 \end{aligned}$$

$\longrightarrow$ Initial Matrix: $\left(\begin{array}{cc|c} \mathbf{1} & -3 & -5 \\ -2 & 7 & 12 \end{array}\right)$

1. **Step 1: Eliminate below the first pivot.**
   $R_2 \leftarrow R_2 + 2R_1$ (Get 0 below the first $\mathbf{1}$)
   $\left(\begin{array}{cc|c} 1 & -3 & -5 \\ 0 & \mathbf{1} & 2 \end{array}\right)$ (Now in Row Echelon Form - REF)
2. **Step 2: Eliminate above the second pivot (Jordan step).**
   $R_1 \leftarrow R_1 + 3R_2$ (Get 0 above the second $\mathbf{1}$)
   $\left(\begin{array}{cc|c} \mathbf{1} & 0 & \mathbf{1} \\ 0 & \mathbf{1} & \mathbf{2} \end{array}\right)$ (Now in Reduced Row Echelon Form - RREF)

Solving a System of Equations

Final Solution

The RREF matrix immediately gives the solution:

$$
\begin{aligned}
x &= 1 \\
y &= 2
\end{aligned}
$$

Application and Solution

Application: Solving Systems of Linear Equations

- The matrix inverse is the most powerful tool for solving a system of linear equations $\mathbf{Ax} = \mathbf{b}$ when a unique solution exists.
- **The Steps:**
  1. Start with the system: $\mathbf{Ax} = \mathbf{b}$
  2. Multiply both sides of the equation on the left by $\mathbf{A}^{-1}$:
  $$\mathbf{A}^{-1}(\mathbf{Ax}) = \mathbf{A}^{-1}\mathbf{b}$$
  3. Since $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$:
  $$\mathbf{Ix} = \mathbf{A}^{-1}\mathbf{b}$$
- **The Final Solution is:**
$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

28. When No Exact Solution Exists

- In Machine Learning, $\mathbf{Ax} = \mathbf{b}$ rarely has an exact solution due to noise/errors.
- We seek the **best approximate solution**.
- This is where the **Projection** concept comes in.

29. Least Squares Approximation

- Goal: Find $\mathbf{x}^*$ that minimizes the error $||\mathbf{A}\mathbf{x} - \mathbf{b}||^2$.
- The solution is given by the **Normal Equation**:

$$\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$$

- **AI Context:** The mathematical foundation for linear regression (finding the "line of best fit").

(Pseudo inverse (A^+))

(Non-singular inverse)

1. Definition and Motivation

When the Standard Inverse Fails

- The *Pseudoinverse (A^+)*, or Moore-Penrose Inverse, is a generalization of the standard inverse (A^{-1}).
- It exists for **any matrix** A:
  - Rectangular matrices ($m \neq n$).
  - Square, but singular (rank-deficient) matrices.
- **Goal:** To find the "best approximate solution," $\mathbf{x}^*$, for the system $Ax \approx b$.

2. Primary Applications of A^+

- **Overdetermined Systems ($m > n$):**
  - More equations than unknowns (e.g., linear regression).
  - No exact solution usually exists.
  - A^+ finds the vector $\mathbf{x}^*$ that minimizes the error $\|Ax - b\|^2$.
  - This is the **Least Squares Solution**.
- **Underdetermined Systems ($m < n$):**
  - Fewer equations than unknowns.
  - Infinitely many solutions exist.
  - A^+ finds the solution $\mathbf{x}^*$ with the minimum length ($\|x^*\|^2$).
  - This is the **Minimum Norm Solution**.

Calculation: Full Rank Cases (One-Sided Inverses)

Simple Formulas when the Rank is Complete

The solution is always $\mathbf{x}^* = \mathbf{A}^+\mathbf{b}$.

Case	Condition	Formula for A^+
Full Column Rank	$\text{rank}(A) = n$ $(m > n)$	$(\mathbf{A}^\mathbf{T}\mathbf{A})^{-1}\mathbf{A}^\mathbf{T}$
Full Row Rank	$\text{rank}(A) = m$ $(m < n)$	$\mathbf{A}^\mathbf{T}(\mathbf{A}\mathbf{A}^\mathbf{T})^{-1}$
Non-Singular	$\text{rank}(A) = m = n$	$\mathbf{A}^{-1}$

Note: These simplified formulas only work if A has full rank.

Full Column Rank Example

Calculating $\mathbf{A}^{+}$ for Least Squares

We use the formula: $\mathbf{A}^{+} = (\mathbf{A}^{\mathbf{T}}\mathbf{A})^{-\mathbf{1}}\mathbf{A}^{\mathbf{T}}$ for the 3×2 matrix A:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (\text{Full Column Rank}, \text{rank} = 2)$$

1. **Calculate $A^T A$:**

$$\mathbf{A}^{\mathbf{T}}\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

2. **Calculate $(A^T A)^{-1}$:** $(\det = 3)$

$$(\mathbf{A}^{\mathbf{T}}\mathbf{A})^{-\mathbf{1}} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

3. **Calculate $\mathbf{A}^{+}$:**

$$\mathbf{A}^{+} = \begin{pmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 1/3 & -1/3 \\ -1/3 & 1/3 & 2/3 \end{pmatrix}$$

Ranked Matercies

For an $m \times n$ matrix A (m rows, n columns), the maximum possible rank is $\min(m, n)$. If a matrix is rank-deficient (i.e., its rank is less than $\min(m, n)$), it means its columns are linearly dependent and its rows are linearly dependent. In this case:

- Both $\mathbf{A^T A}$ and $\mathbf{A A^T}$ will be singular matrices.
- The inverses $(\mathbf{A^T A})^{-1}$ and $(\mathbf{A A^T})^{-1}$ will not exist.